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Kinetics of air mixing–anaerobic digestion: High organic substrate concentrations enhance system resistance to oxygen stress

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ABSTRACT

Air mixing–anaerobic digestion is proposed to overcome the limitations of traditional mixing methods, which can increase methane output and reduce H_2S content. However, the effects of air mixing can be either inhibitory or facilitative, depending on the duration of oxygen stress and the elevated methane production rate. Kinetics models were developed to describe the variations (increase and decrease) in ORP and methane production. The results revealed that the increase in ORP was negatively correlated with the organic substrate concentration, but positively correlated with mixing intensity. The decrease in ORP showed a positive correlation with both the organic substrate concentration and the maximum ORP reached at the end of air mixing. The maximum rate of ORP increase and decrease were 66.0 mV/min and 3.03 mV/min, respectively. After air mixing stopped, when ORP decreased below −370 mV (0 mg/L DO), the methane production increased. The maximum methane production could be elevated to 358 mL/h at sufficient substrate. High organic substrate concentrations are critical for system resistance to oxygen-induced inhibition. This resistance is primarily achieved through accelerated oxygen consumption and stabilized anaerobic environments for methane production. The developed methane output models enable the quantitative prediction of methane output and offer guidance for optimizing air mixing parameters.

1. Introduction

The anaerobic digestion technology is an effective strategy for waste treatment and utilization, with the advantages of less energy consumption, a high organic loading rate, and the ability to recover a clean energy source [1,2]. Proper mixing is the key to improving the efficiency of anaerobic digestion with high concentrations of suspended solids [3,4]. Mechanical mixing in the traditional method is characterized by high energy consumption, and difficult maintenance [5]. Biogas mixing has the advantages of simple maintenance and low equipment costs [6]. However, for biogas mixing, there are problems such as safety hazards and biogas shortages at start up phase [7]. To overcome these issues, our team proposed a novel strategy of using air to replace biogas as a carrier to drive mixing, and establish an innovative technology called air mixing–anaerobic digestion [8,9]. Compared with mechanical mixing and biogas mixing, this technology was able to increase volumetric methane production by 6.4%–19.6%, and it also provides the added benefit of reducing H_2S in the biogas. Air mixing–anaerobic digestion has already been applied in full–scale biogas plants and has achieved superior performance [10], primarily because air mixing not only improves the mass transfer efficiency between microorganisms and the substrate (with a relative contribution of 62.9%) but also creates a micro–oxygen environment (with a relative contribution of 37.1%) [4,11]. Even though the micro–oxygen environment can effectively promote anaerobic digestion, thereby resulting in increased methane production [12,13]. Inappropriate air mixing intensity and duration can lead to a decrease in the normal methane production time, thus leading to a reduction in the total methane output [10]. The effects of air mixing on the growth and metabolism of microorganisms and the performance of methane production are clearly complex processes. To maintain efficient anaerobic digestion under air mixing condition, parameters such as intensity, time, and frequency must be precisely regulated. This requires the development of kinetic models capable of optimizing these key mixing parameters.

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Kinetic models have been widely used in studies of anaerobic digestion to describe biogas production, as well as to predict and regulate operational performance [14]. Kinetic models include mainly early experience models and process models. Many experience models have been developed such as the first–order kinetic model and the modified Gompertz model [15]. Process models that are based on biochemical reaction mechanisms include the Monod model [16], the Contois model [17], the Stovere Kincannon model [18], and the Chen–Hashimoto model [19,20] etc. In recent years, the kinetics of anaerobic digestion in a micro–oxygen environment have gradually attracted increasing attention [21]. The role of dissolved oxygen (DO) in various stages of anaerobic digestion was characterized by ADM1 [22]. However, only simple substrates such as glucose and primarily substrate degradation have been considered [23], and the effects of DO dynamics on methane production and ORP variation have been neglected. Moreover, the complexity of ADM1 limits its applicability in anaerobic digestion with complex substrates or actual plants. In particular, the ADM1-Ox (oxygen) model incorporates 25 state variables and 22 biochemical processes, making it difficult to obtain all required state variables in practical anaerobic digestion (biogas) plant [22]. Furthermore, due to the inherent complexity of anaerobic digestion systems, accurate measurement of DO also presents considerable challenges.

Although air mixing–anaerobic digestion has been established, its optimization still relies on the development of appropriate kinetic models. To clarify the quantitative effects of air mixing on the ORP and methane production. Air mixing–anaerobic digestion experiments were carried out by using an anaerobic sequencing batch reactor (ASBR). Based on these experiments, kinetic models describing ORP variation (increase and decrease) and methane production were developed. Moreover, a integrated model for determining the mixing intensity, time, and frequency of air mixing–anaerobic digestion was established. These models developed in this study can lay the foundation for advancing air mixing–anaerobic digestion technology and challenge the conventional understanding of oxygen's role in anaerobic processes. Further investigation of oxygen-driven kinetics will not only fill the existing knowledge gap regarding the quantitative effects of oxygen on anaerobic digestion but also provide a theoretical basis for improving and optimizing this emerging air-mixing technology. Moreover, the findings also can offer new insights into methane emissions under microoxic and anaerobic conditions.

2. Materials and methods

2.1. Laboratory–scale reactor, feedstock and inoculum

The reactor was an ASBR system with a diameter of 11 cm, length of 45 cm, total volume of 4.5 L, and working volume of 4 L. It was equipped with an outer jacket, as well as an inlet and outlet for circulating heated water. Temperature was maintained at 35 °C by pumping hot water from a thermostatic bath through the jacket. Air for mixing was injected into the reactor via an air pump and an aeration tray at the bottom of the reactor. The reactor was equipped with pH, DO, and ORP probes connected to a data recorder (M300, Mettler Toledo, Urdof, Switzerland).

The swine wastewater used as organic substrate of anaerobic digestion was obtained from a farm in Jianyang City, Sichuan Province, China. A 0.85 mm (20 mesh) sieve was used to remove the large suspended substance. The filtrate was stored at 4 °C. Before use, the filtrate was diluted with tap water to adjust the chemical oxygen demand (COD) concentration to the required level. The inoculum sludge was obtained from a mesophilic anaerobic digestion reactor treating swine manure. A detailed characterization of the swine wastewater and anaerobic inoculum sludge is shown in Table S1.

2.2. Experimental design

At the start of the experiment, 1.5 L of anaerobic inoculum sludge and 2.5 L of substrate were added to each of three identical reactors. The reactors were operated in an ASBR mode. Each daily cycle consisted of withdrawal, feeding, and reaction stages. At a fixed time each day, 1 L of digestate was first withdrawn and then fed with 1 L of swine wastewater prepared at the designated organic concentration. When the daily biogas production varied within 10% of the average value over a period of 1–2 weeks, the reactor was considered to have reached steady state conditions, and the experiment could be carried out.

2.2.1. Experimental design for the response of the ORP to DO variation in air mixing–anaerobic digestion

The swine wastewater used as organic substrate in the experiment was diluted so that the substrate concentration was 10 g COD/L. First, the organic substrate was added to the ASBR. After the ORP stabilized at approximately −450 to −460 mV, air mixing began at an intensity of 66.7 mL air/min. The DO and ORP values were both recorded every minute with a data recorder. Mixing continued for 30 min, and data recording was stopped when the DO and ORP stabilized.

2.2.2. The effect of air mixing on ORP variation and methane production

After the ORP stabilized (at approximately −450 to −460 mV), the M300 recorder was started. The ORP values were recorded at 1 min intervals until the initial ORP recovered to the background values at the end. For each of the 5 organic substrate concentrations (2, 4, 6, 8, and 10 g COD/L), air was pumped into the mixed liquid at flow rates (mixing intensities) of 0, 33.3, 66.7, and 100 mL air/min for 120 min. The biogas output and CH_4 concentration were measured every 1 h for 24 h after the air mixing stopped. At the end of each experiment, the anaerobic inoculum sludge which was also originated from the same anaerobic digestion reactor, was replaced the sludge of the previous experiments to ensure that the sludge activity stabilized before the next experiment started.

2.3. Analysis methods

Total solid (TS), volatile suspended solid (VS), and suspended solid (SS) contents were determined by the weighing method [24]. Samples were dried at 105 °C to constant weight using a DHG-9075 A drying oven (Yiheng, China), followed by ignition at 550 °C in a KSL-1100× muffle furnace (Hefei kejing, China). ORP, DO, and pH were measured using Mettler-Toledo electrodes, including an InLab Redox Pro ORP electrode (Mettler-Toledo, Switzerland), an InLab OptiOx optical DO sensor (Mettler-Toledo, Switzerland), and an InLab Expert Pro pH electrode (Mettler-Toledo, Switzerland). The COD measured in this study was primarily the TCOD (total chemical oxygen demand) of swine wastewater. Analyses were performed using the potassium dichromate digestion method with instruments manufactured by LH-A230 (Lianhua Technology, China). Gas samples (approximately 5 mL) were collected from the anaerobic digestion reactors to determine the biogas composition by gas chromatography (model SP–2100 A, TCD detection, Beijing Beifen Ruili Analytical Instrument Co., Ltd.) with He as the carrier gas, and the temperatures of the column, oven, and temperature detector were 60 °C, 90 °C, and 105 °C, respectively. The methane production was determined by multiplying the volume of biogas by the methane content in the biogas at hourly intervals.

2.4. Model fitting

The experimental data from all reactors were evaluated for model fitting using OriginPro 2021 (OriginLab Corporation, USA), and the correlation coefficients (R^2) were calculated accordingly. For kinetic derivation and subsequent analysis, MATLAB R2022b (MathWorks, USA) was employed to construct the mathematical model, generate surface plots incorporating the experimental datasets, and perform statistical parameter fitting.

3. Results and discussion

3.1. Kinetics of ORP variation in air mixing–anaerobic digestion

3.1.1. Response of the ORP to DO variation

OPR and DO electrodes are monitoring devices widely used in wastewater treatment. DO diffuses rapidly in liquor and is immediately consumed by microorganisms and reducing compounds. As a result, the oxygen concentration at the electrode surface changes instantaneously. This makes it difficult for electrochemical sensors to obtain stable signals at very low concentrations, resulting in a relatively high detection limit for DO electrodes. Because ORP responds sensitively and monotonically to oxygen intrusion, while DO is rapidly depleted and difficult to measure at low levels [25,26]. Therefore, ORP can effectively replace DO as a more stable indicator of micro-oxygen conditions [27].

The changes in the DO and ORP in the mixed liquid were measured in air mixing–anaerobic digestion (Fig. 1a). When mixing began, the changes in the DO and ORP in the mixed liquid exhibited similar trends. Both the ORP and DO increased rapidly and then gradually stabilized. Origin software was used to determine the quantitative relationship between DO and ORP by fitting the data to an exponential model (Eq. (1)); the response parameters are shown in Table 1.

Table 1
Parameters of the ORP and Ln (DO) response model.

Raw material	ORP_0 (mV)	k (mV)	R^2
Swine wastewater	−111	82.2	0.961

$$ORP_s = k^* \mathrm{Ln}\,(C_{DO}) + ORP_0 \quad (1)$$

In this equation, C_{DO} is the DO content of the mixed liquid at a certain time (mg/L); ORP_s is the ORP value of the mixed liquid at the same time (mV); k is the rate of the response of ORP (mV) to the change in Ln (DO) variation; and ORP_0 is the ORP value at a DO concentration of 1 mg/L.

This model not only provides a simpler method to determine the DO content, and practical approach for estimating DO in air mixing–anaerobic digestion, but also enables the inference of low DO concentrations from ORP measurements. This capability provides a foundation for further elucidating how oxygen dynamics influence anaerobic digestion processes.

3.1.2. Kinetics of the increase in the ORP in air mixing–anaerobic digestion

The effects of the substrate concentration and air mixing intensity on

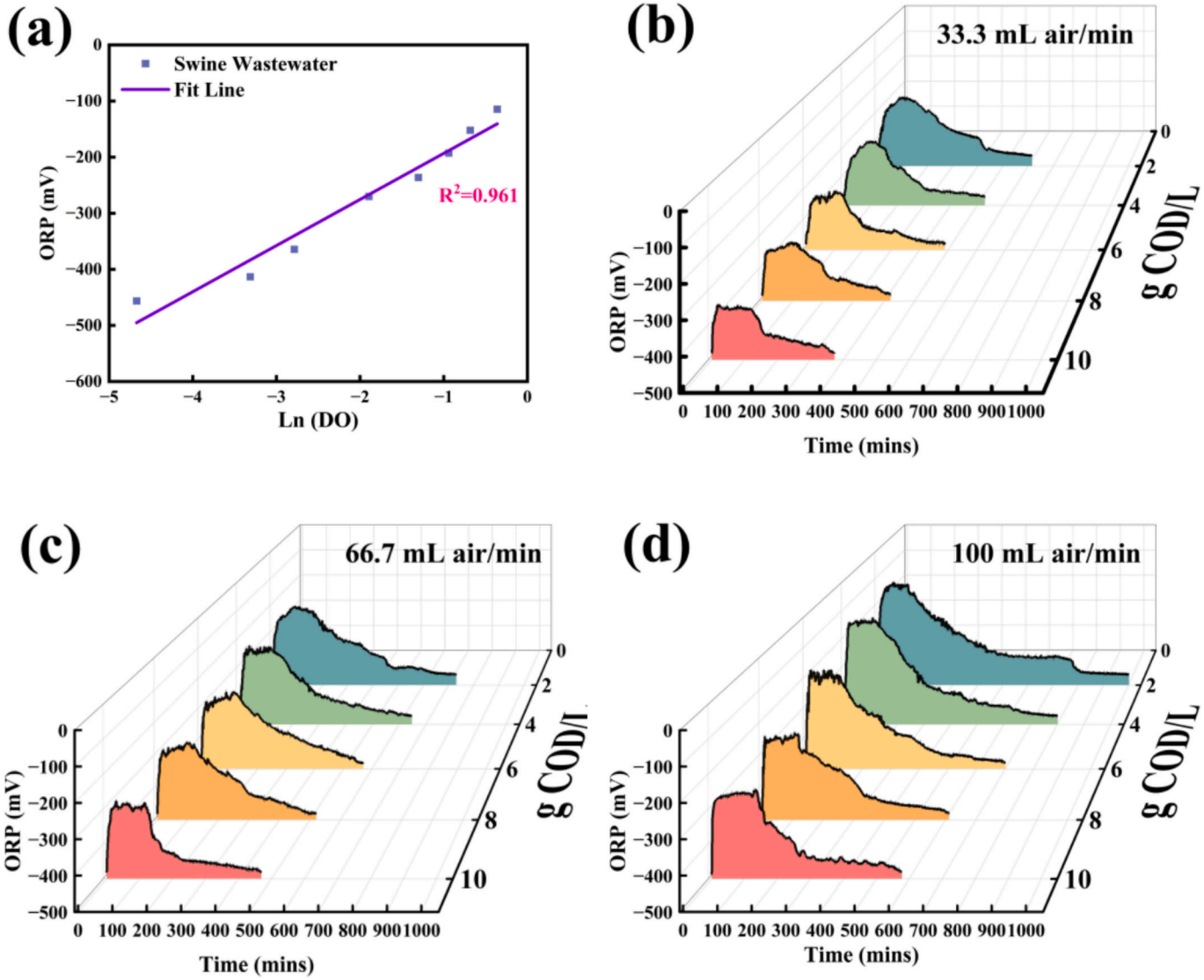


Fig. 1. Response of the ORP to Ln(DO) variation in air mixing–anaerobic digestion (a), ORP variations at different organic substrate concentrations under air-mixing intensities of 33.3 mL air/min (b), 66.7 mL air/min (c), and 100 mL air/min (d).

the ORP were showed in Fig. 1b–c. When air mixing started, the ORP increased rapidly due to the increase in the DO content in the mixed liquid. The DO content subsequently no longer increased consistently because of oxygen consumption by facultative anaerobes. With the balance of oxygen dissolution and oxygen consumption, the DO in the mixed liquid tended to stabilize. A similar trend was identified in aerobic aeration tanks [28]. The oxidation–reduction reaction also gradually became constant, and the ORP tended to be stable. Notably, the ORP increased to the maximum ORP value (ORP_{max}) that could be reached in the mixed liquid at a given substrate concentration with increasing mixing intensity. The ORP change process and the ORP_{max} reached in air mixing were recorded (the range of the ORP value change and the time needed are presented in Table S2).

During 120 min of air mixing–anaerobic digestion, the increase in the ORP was influenced by both the organic substrate concentration and the air mixing intensity. For example, when the substrate concentration was 2 g COD/L, ORP_{max} was −244 mV at a mixing intensity of 33.3 mL air/min. However, at a mixing intensity of 66.7 mL air/min, ORP_{max} reached −219 mV. At 100 mL air/min, ORP_{max} further increased to −120 mV. Under the same mixing intensity, higher substrate concentrations led to faster oxygen consumption by microbes [13]. Therefore, the increase in the organic substrate concentration reduced ORP_{max}. Specifically, when the mixing intensity was 33.3 mL air/min, ORP_{max} was −120 mV at 2 g COD/L, −136 mV at 4 g COD/L, −195 mV at 6 g COD/L, and − 237 mV at 8 g COD/L. At 10 g COD/L, ORP_{max} decreased to −357.4 mV. The intensity of air mixing and the organic substrate concentration exerted combined effects on oxygen dissolution and consumption, which consequently changed the maximum DO and ORP that the mixed liquid could achieve.

The rate of increase in the ORP (R_{ORP}) was calculated, as shown in Table S3. Improving the air mixing intensity significantly increased the oxygen dissolution rate. Molecular oxygen has a relatively high ORP value (+0.86 V) and can rapidly accept electrons [29]. A low air mixing intensity (33.3 mL air/min) brought the mixed liquid with 2–4 g COD/L into only a micro–oxygen environment (−300 to −150 mV) [12], and the R_{ORP} was relatively slow (6.08–6.72 mV/min). However, when the intensity was increased to 100 mL air/min at 2–4 g COD/L, the mixed liquid directly entered the aerobic environment (> − 150 mv, −120 ~ −136 mV). The R_{ORP} also increased significantly to 15.9–17.1 mV/min. The DO at a high concentration consumed electrons, which caused a rapid increase in the ORP. In addition, at the same air mixing intensity of 100 mL air/min, the R_{ORP} was only 10.9 mV/min when the organic substrate concentration was increased to 10 g COD/L. The R_{ORP} decreased by a maximum of 36.3% at 10 g COD/L compared with that at 2 g COD/L. The high organic concentration of the substrate clearly resisted the air mixing–anaerobic digestion system enter the aerobic environment.

Both R_{ORP} and ORP_{max} was affected by two factors (the organic substrate concentration and mixing intensity). The Michaelis–Menten reaction model of two substrates in random order is widely used to describe the kinetics of anaerobic digestion [30,31]. A kinetic model of the increase in ORP (Eq. (2)) and the maximum ORP (ORP_{max}) (Eq. (3)) in air mixing–anaerobic digestion was developed. The experimental results (Tables S2–3) were used to fit the equations, and the kinetic parameters (Tables 2 and 3) and simulation results (Fig. 2) were obtained.

$$R_{ORP} = P_m * \left(1 - \frac{S}{k_s + S}\right) * \frac{M}{k_m + M} \quad (2)$$

Table 2
Kinetic parameters for the rate of increase in the ORP.

	P_m (mV/min)	k_S (g COD/L)	k_m (mL air/min)	R^2
R_{ORP} (mV/min)	66.0	17.1	258	0.943

Table 3
Kinetic parameters for the maximum ORP.

	ORP_{bk} (mV)	k_S (g COD/L)	k_m (mL air/min)	R^2
ORP_{max} (mV)	−370	19.0	75.5	0.921

where R_{ORP} is the increase rate of the ORP (mV/min), P_m is the maximum increase rate of the ORP (mV/min), S is the organic substrate concentration (g COD/L), M is the air mixing intensity (mL air/min), k_s is the half–saturation constant based on the organic substrate concentration S (g COD/L), and k_m is the half–saturation constant based on the mixing intensity M (mL air/min).

$$ORP_{max} = ORP_{bk} * \left(1 + \frac{S}{k_s + S}\right) * \left(1 - \frac{M}{k_m + M}\right) \quad (3)$$

where ORP_{max} is the maximum ORP that the mixed liquid can reach (mV), ORP_{bk} is the ORP value at 0 g COD/L organic substrate and 0 mL air/min mixing intensity (mV), S is the organic substrate concentration (g COD/L), M is the air mixing intensity (mL air/min), k_s is the half–saturation constant based on the organic substrate concentration S (g COD/L), and k_m is the half–saturation constant based on the mixing intensity M (mL air/min).

The coefficient of determination R^2 indicated that the two models could better reflect the effects of the organic substrate concentration and air mixing intensity on the increase in ORP. P_m represents the maximum rate of R_{ORP} (66.0 mV/min) that can be achieved in the absence of organic substrate when the air mixing intensity is much greater than the half–saturation constant k_m (258 mL air/min). Moreover, under high–intensity air mixing, the methane production rate can be reduced to half of P_m as the organic substrate concentration gradually increases to reach the half–saturation constant k_s (17.1 g/L).

In addition, assuming that there is no influence from air mixing, the system should be in a completely anaerobic environment in the kinetic model of ORP_{max}. ORP_{bk} (−370 mV) was calculated using Eq. (3), which is consistent with the ORP range of the anaerobic environment in anaerobic digestion. Liu et al. [32] concluded that the suitable range for methanogens is −350 mV or even lower. The ORP_{bk} (−370 mV) can be regarded as the ORP value that corresponds to the anaerobic environment that was suitable for methane production when swine wastewater was used as the substrate in this study. Moreover, even with a relatively high organic substrate concentration, air mixing could have caused rapid entry into the aerobic environment (> − 150 mV) when the intensity approached the half–saturation constant k_m (75.5 mL air/min). Although hydrolytic and acidogenic bacteria were not affected by the aerobic environment, the methanogens were significantly inhibited [23]. The period in which the ORP was greater than the ORP_{bk} was detrimental to methane production.

3.1.3. Kinetics of the decrease in ORP in air mixing–anaerobic digestion

After air mixing stopped, oxygen was gradually consumed by the microorganisms. The oxidizing substances in the mixed liquid were reduced, and the ORP also gradually decreased. Lim et al. [33] found that increasing the substrate concentration improved the rate of oxygen consumption by microorganisms. This finally shortened the time required for the ORP to return to an anaerobic environment (−300 mV). For example, in anaerobic digestion with 2 g COD/L, at a mixing intensity of 100 mL air/min, recovery to the anaerobic environment took 181 min. However, for the sample with 10 g COD/L, recovery took only 28 min. The rapid recovery of the ORP demonstrated that the rate of oxygen consumption increased. Together, the DO and organic substrate concentrations influenced the rate of decrease in the ORP. The rates of increase and decrease in the ORP were analysed quantitatively by kinetics.

After air mixing stopped, the ORP in the mixed liquid began to decrease gradually and eventually reached an anaerobic environment

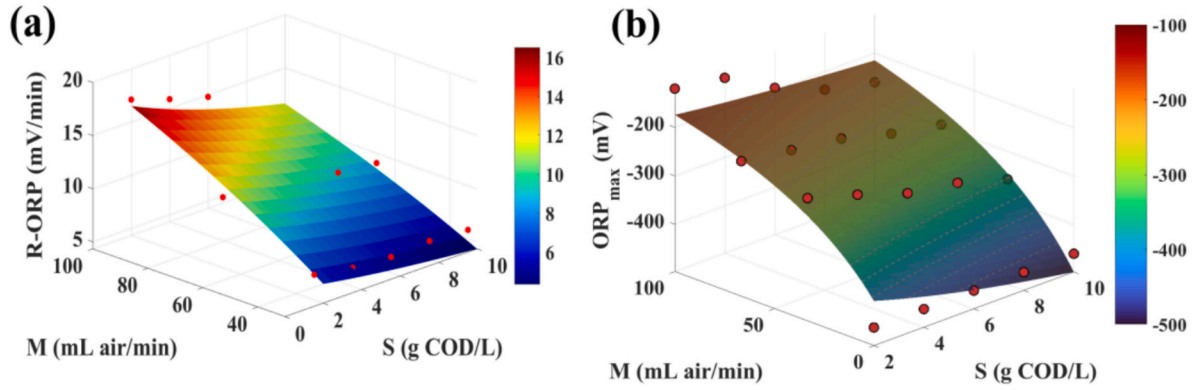


Fig. 2. Effects of the organic substrate concentration and air mixing intensity on the rate of increase in the ORP (a) and the maximum ORP (b).

from an aerobic or micro–oxygen environment. By analyzing the time required for the ORP to return to an anaerobic environment, the rate of decrease in the ORP (U_{ORP}) was calculated (Table S3). The results revealed a positive correlation between the organic substrate concentration and U_{ORP}. Previous studies have shown that a high organic substrate concentration promotes an increase in the oxygen consumption rate by microorganisms and generates more reducing substances with low electric potentials [34–36]. The U_{ORP} was approximately 0.595–0.993 mV/min at 2 g COD/L, whereas the U_{ORP} was 1.43–1.46 mV/min at 10 g COD/L. At a mixing intensity of 100 mL air/min, the rate of decrease in the ORP at substrate of 10 g COD/L was approximately 1.45–fold of that at 2 g COD/L.

Both substrate metabolism and respiration can be used by microorganisms to produce energy. The level of DO also determines the rate of change in U_{ORP}. As a correlation exists between the DO content and the ORP in anaerobic digestion (Eq. (1)), the DO value used in this study refers specifically to the initial DO concentration at the onset of ORP decrease. This initial DO content was calculated from the maximum ORP (ORP_{max}) reached at the end of air mixing (Table S2). The above results can be explained by a kinetic model developed as in Eq. (3). The model reflects the simultaneous influence of two substrates (organic substrate concentration and DO content) on the rate of DO consumption (ORP decrease) by microorganisms. The experimental data were fitted to Eq. (4) to obtain the kinetic parameters (Table 4, Fig. 3a) for the decrease in ORP.

$$U_{ORP} = U_m * \frac{S}{k_s + S} * \frac{I}{k_I + I} \quad (4)$$

where U_{ORP} is the decrease rate of the ORP after air mixing stops (mV/min), U_m is the maximum ORP decrease rate (mV/min), S is the organic substrate concentration (g COD/L), I is the initial DO content at the time when air mixing is stopped (mg/L), k_s is the half–saturation constant based on the organic substrate concentration S (g COD/L), and k_I is the half–saturation constant based on the DO concentration (mg/L).

In this kinetic model, U_m indicates the maximum ORP decrease rate (3.03 mV/min), which can be achieved with a high organic substrate concentration and a high DO content. By this calculation, anaerobic digestion was estimated to be able to move from a micro–oxygen environment to an anaerobic environment (from −150 to −300 mV) in approximately 50 min. Furthermore, k_S (5.52 g COD/L) and k_I (0.056 mg/L) were able to reflect the extent of the effect of these substrates on U_{ORP}. The organic substrate concentration and DO content both have pronounced effects on U_{ORP}. The oxygen consumption rate during the decrease in the ORP is correlated with the abundance and activity of microorganisms. High DO induces anaerobic microorganisms to increase the expression of functional genes in response to oxidative stress [37]. Some anaerobic microorganisms even use the same mechanism to consume oxygen [38], which mitigates the damaging effect on anaerobic microorganisms. Moreover, a high organic substrate concentration provides sufficient nutrients for bacteria to ensure normal growth, which is conducive to accelerating facultative anaerobic respiration and increasing the rate of oxygen consumption.

Table 4
Kinetics parameters for the decrease in the ORP.

	U_m (mV/min)	k_S (g COD/L)	k_I (mg/L)	R^2
U_{ORP}	3.03	5.52	0.056	0.927

3.2. Kinetics of methane production in air mixing–anaerobic digestion

The CH_4 content (Fig. 4a–c) and biogas output (Fig. S1a–d) were recorded, which were used to estimate the methane output (Fig. 4e–h) and methane production rate (Table S4). The results revealed that the methane production process can be divided into three phases (abnormal, promotion, and normal) (Fig. 4d). The methane production rates in the three stages can be designated as R_i, R_e, and R_{ck}. In the abnormal phase, the methane concentration in the biogas is lower than that in the control (no air mixing). The times of abnormal phase includes the air mixing time and the recovery time. In the promotion phase, the methane production rate increased due to the enhanced mass transfer efficiency and the creation of a micro–oxygen environment by air mixing. After the mixing effect is mitigated, methane production enters the third phase, namely, the normal phase. At this phase, the ORP returned to a reducing environment, the DO was completely consumed by microorganisms, and the rate of methane production was the same as that of the control.

3.2.1. Methane production in abnormal phase with oxygen inhibition

The CH_4 concentrations were extremely low (less than 1%) during air mixing. After the air mixing stopped, the CH_4 concentration gradually recovered to a normal level (more than 40%). However, the methane production rate was significantly lower than that of the control under the same organic substrate concentration during this abnormal phase. For example, at 10 g COD/L, the control reactor exhibited a methane production rate of 70.1 mL/h, whereas even at the lowest air mixing intensity of 33.3 mL air/min, the rate decreased markedly to 41.1 mL/h, which is far lower than that of the control. Meanwhile, organic substrate concentration have a substantial impact on the methane production rate in the abnormal phase. Under an air mixing intensity of 33.3 mL air/min, the methane production rate was 7.10 mL/h at a COD concentration of 2 g/L. When the COD concentration increased to 10 g/L, the methane production rate rose to 41.1 mL/h, representing approximately a 4.79-fold increase. Similarly, at an air mixing intensity of 100 mL/min, the methane production rate at 10 g COD/L (37.1 mL/h) was approximately 10.8 times greater of than that at 2 g COD/L (3.15 mL/h). The methane production rate increases with increasing organic substrate concentration in the micro–oxygen environment of anaerobic digestion

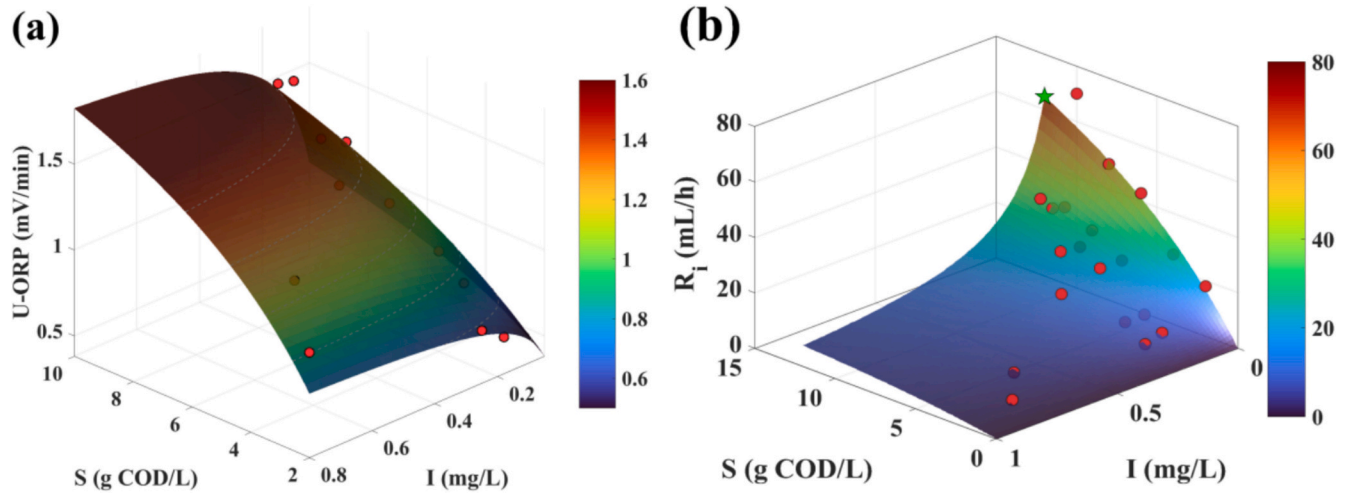


Fig. 3. Effects of organic substrate concentration and DO content on the decrease in ORP (a), and methane production rate during recovery (b).

[21]. Higher substrate concentrations improved the ability to withstand oxygen-induced stress in air mixing–anaerobic digestion system.

The methanogen activity was also significantly affected by the DO content. This observation is consistent with previous findings of a significant negative correlation between the DO content and methane production [39]. At 2 g COD/L, the methane production rate decreased from 7.10 mL/h at 33.3 mL air/min (DO, 0.181 mg/L) to 3.15 mL/h at 100 mL air/min (DO, 0.046 mg/L).

The methane production rate clearly depends on the DO and organic substrate concentrations in the abnormal phase, as shown in Fig. 3b. There is substrate competition between facultative anaerobes and methanogens. A kinetic model of methane production under oxygen inhibition was developed on the basis of the double–substrate kinetic model (Eq. (5)) in the abnormal phase. The kinetic parameters are shown in Table 5.

$$R_i = R_{im} * \frac{S}{k_s + S} * \left(1 - \frac{I}{k_I + I}\right) \quad (5)$$

where R_i is the methane production rate in the abnormal phase (mL/h), S is the substrate concentration (g COD/L), I is the DO content at the time when air mixing is stopped (mg/L), R_{im} is the maximum methane production rate (mL/h), k_s is the half–saturation constant based on the substrate concentration S (g COD/L), k_I is the half–inhibition constant based on the DO concentration I (mg/L), and R_{ck} is the methane production rate in the control when I is 0 mg/L.

When the DO concentration reached k_I (0.132 mg/L), the methane production rate was reduced to 50% of the maximum methane production rate (R_{im}, 152 mL/h) because of the inhibition of DO. The k_I value can also somewhat reflect the degree of sensitivity of the methanogens to oxygen. The lower k_I is, the more intense the response of methanogens to DO. Trace amounts of oxygen are then able to lead to an increase in ORP, thereby significantly inhibiting the activity of the methanogens and decreasing the rate of methane production [40]. When the DO content is much greater than k_I, the methane production rate is completely controlled by DO. Seuntjens et al. [41] found that methane production is easily and strongly inhibited by DO, and increasing methane production requires a higher organic substrate concentration. As the organic substrate concentration approaches k_S (16.2 g COD/L), the methane production rate can reach 50% of the maximum methane production rate. When the substrate concentration is much greater than k_S and the DO concentration approaches 0 mg/L, the methane production rate approaches the maximum methane production rate. Thus, this competitive inhibition kinetic model can be used to understand the effects of the organic substrate and DO (mixing intensity) on the methane production rate. In addition, in the control (no air mixing), the DO concentration was 0 mg/L. At that time, the methane production rate R_{ck} was related only to the organic substrate concentration.

3.2.2. Methane production in promotion phase

After air mixing stopped, the methanogens gradually recovered normal activity [42] the methane concentration recovered, and entered the promotion phase of methane production. The results revealed that low–intensity air mixing and high organic substrate concentrations could increase the methane production rate (Table S4). Specifically, at 10 g COD/L, the methane production rate increased to 97.9 mL/h at 33.3 mL air/min, which was significantly higher than that of the control (70.1 mL/h). A micro–oxygen environment can promote hydrolysis and mitigate acid accumulation, thereby resulting in an increased rate of methane production [43]. Notably, the results revealed that the process of promoting the methane production rate lasted approximately 5 h (promotion phase), as Fig. 4d showed. The rate of methane production subsequently decreased to the level of the control (normal phase). This phenomenon has been similarly observed in previous studies [10]. In the promotion phase, DO is consumed by facultative anaerobes. The methane production rate gradually increased. The methane production rate was related only to the substrate concentration. The model can be simplified from Eq. (5) to Eq. (6).

$$R_e = R_{em} * \frac{S}{k_s + S} \quad (6)$$

where R_e is the methane production rate in the promotion phase (mL/h), S is the organic substrate concentration (g COD/L), R_{em} is the maximum methane production rate (mL/h), and k_s is the half–saturation constant based on the organic substrate concentration S (g COD/L).

The kinetic parameters (Table 6) indicated that half–saturation based on the organic substrate concentration (k_s of 32.6 g COD/L) is an important factor that influences the methane production rate. Theoretically, the methane production rate can reach 358 mL/h (R_{em}) for a sufficiently high substrate concentration (much higher than 32.6 g COD/L).

3.3. Model of methane output in air mixing–anaerobic digestion

The calculation of the methane output (Q) of air mixing–anaerobic digestion in a day (24 h) includes three phases. First, during the air mixing time, the methane output is negligible because of the low methane concentration. However, the mixing time (t_m) needs to be accounted for. After air mixing stops, the physiological activity of the methanogens gradually recovers. The time required for the ORP

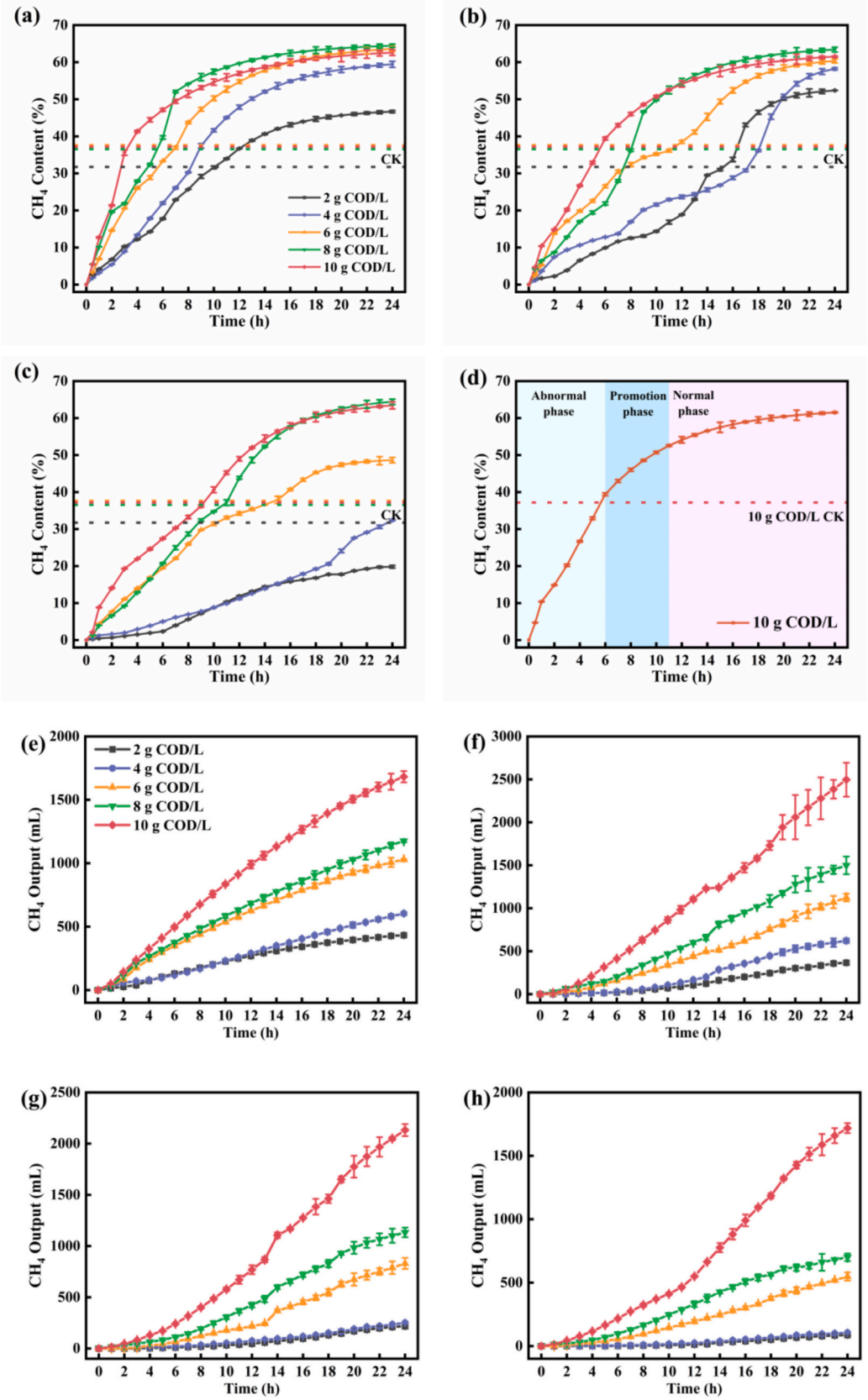


Fig. 4. Recovery of CH_4 content under different organic substrate concentrations at 33.3 mL air/min (a), 66.7 mL air/min (b), and 100 mL air/min (c). the abnormal, promotion, and normal phases at 10 g COD/L under an air mixing intensity of 66.7 mL air/min (d). Variations in CH_4 output for the control (CK) (i) and at air mixing intensities of 33.3 mL air/min (j) 66.7 mL air/min (k), and 100 mL air/min (l) at different organic substrate concentrations.

Table 5
Kinetic parameters of methane production in the abnormal phase.

	R_{im} (mL/h)	k_s (g COD/L)	k_I (mg/L)	R^2
CH_4 production rate	152	16.2	0.132	0.912

Table 6
Kinetic parameters of methane production in the promotion phase.

	R_{em} (mL/h)	k_s (g COD/L)	R^2
R_e	358	32.6	0.967

(ORP_{max}) to recover to below −370 mV was defined as the recovery time within the abnormal phase (t_r). By using a kinetics model of the maximum ORP (ORP_{max}) and the decrease rate in ORP (U_{ORP}), t_r can be calculated. t_m and t_r make up the abnormal phase (t_i).

$$\text{The time of abnormal phase } (t_i) = \text{Air mixing time } (t_m) + \text{recovery time } (t_r)$$

$$= t_m + (ORP_{max} - (-370))/U_{ORP} \quad (7)$$

When the system subsequently returns to an anaerobic state with ORP values less than or equal to −370 mV (anaerobic environment, from Eq. (4)), the methane concentration can increase to above 40%. The time when the methane production rate is higher than that of the control can be defined as the promotion phase (t_e). By combining the mixing frequency (f), the remaining time is the normal phase (t_n, Eq. (9)).

$$\text{The time of promotion and normal phases } (t_e) = 24 - f^* t_i/60$$
$$= 24 - f(t_m + t_r)/60$$
$$= 24 - f^* t_m/60 - f(ORP_{max} - (-370))/U_{ORP}/60 \quad (8)$$

$$\text{The time of normal phase } (t_n) = 24 - f^*(t_m + t_r)/60 - 5f$$
$$= 24 - f^* t_m/60 - f(ORP_{max} - (-370))/U_{ORP}/60 - 5f \quad (9)$$

By combining the methane production rates and the time of the abnormal phase, promotion phase and normal phase, and the mixing frequency, a model for the calculation of methane output (Q) in air mixing–anaerobic digestion can be derived (Eq. (10)).

$$Q = f^* t_r^* R_i + 5f^* R_e + t_n^* R_{ck}$$
$$= f^*[(ORP_{max} - (-370))/U_{ORP}]^* R_i + 5f^* R_e + [24 - f^* t_m$$
$$- f^*(ORP_{max} - (-370))/U_{ORP} - 5f]^* R_{ck} \quad (10)$$

where Q is the methane output in air mixing–anaerobic digestion (mL/d), f is the mixing frequency (times/d), t_m is the time of air mixing (min), t_i is the time of methane production in the abnormal phase (min), t_e is the time in promotion and normal phases (min), R_i is the methane production rate in the normal phase (mL/(h)) (from Eq. (5)), R_{ck} is the methane production rate in the normal phase (mL/(h)) (from Eq. (5)), R_e is the methane production rate in the promotion phase (mL/(h)) (from Eq. (6)), ORP_{max} is the ORP when air mixing is stopped (mV) (from Eq. (3)), and U_{ORP} is the rate of ORP decrease after air mixing is stopped (mV/min) (from Eq. (4)).

The equations of methane production rates (R_i, R_e and R_{ck}) and the time (t_r and t_n) are further brought into the Q model.

$$Q = f^* \left(\frac{ORP_{bk} * \left(1 + \frac{S}{k_s+S}\right) * \left(1 - \frac{M}{k_m+M}\right) - (-370)}{U_m * \frac{S}{k_s+S} * \frac{I}{k_I+I}} \right) \Big/ 60 * R_i * \frac{S}{k_s+S} * \left(1 - \frac{I}{k_I+I}\right) + f^* 5 * R_e * \frac{S}{k_s+S}$$
$$+ \left(24 - f^* t_m - f^* \frac{ORP_{bk} * \left(1 + \frac{S}{k_s+S}\right) * \left(1 - \frac{M}{k_m+M}\right) - (-370)}{U_m * \frac{S}{k_s+S} * \frac{I}{k_I+I}} - f^* 5 \right) \Big/ 60 * R_{ck} * \frac{S}{k_s+S} \quad (11)$$

where, M is the air mixing intensity (mL air/min), I is the DO content at the time when air mixing is stopped (mg/L), S is the organic substrate concentration (g COD/L).

Finally, the final quantitative model of methane output (Eq. (12)) in air mixing-anaerobic digestion was obtained by combining the kinetics constants gained in this study.

$$Q = f^* \left(\frac{-370 * \left(1 + \frac{S}{19.0+S}\right) * \left(1 - \frac{M}{75.5+M}\right) - (-370)}{3.03 * \frac{S}{5.52+S} * \frac{I}{0.056+I}} * \frac{1}{60} \right) * 152 * \frac{S}{16.2+S} * \left(1 - \frac{I}{0.132+I}\right) + 5 * f * 358 * \frac{S}{32.6+S}$$
$$+ \left(24 - f^* t_m - 5^* f - f^* \frac{-370 * \left(1 + \frac{S}{19.0+S}\right) * \left(1 - \frac{M}{75.5+M}\right) - (-370)}{3.03 * \frac{S}{5.52+S} * \frac{I}{0.056+I}} * \frac{1}{60} \right) * 152 * \frac{S}{16.2+S} \quad (12)$$

Note: As the mixing frequency and mixing time increase, the promotion phase and the normal phase are shortened or even absent.

3.4. Validation and prediction of the methane output model

To validate the model, an independent experimental was conducted following the procedures described in Text S1. By using swine wastewater at 5 g COD/L (Fig. 5a), the model demonstrated strong predictive accuracy. Under both control condition without air mixing and the three air mixing intensities (33.3, 66.7, and 100 mL air/min), the relative errors remained below 10%, ranging from 1.58% to 5.25%.

Subsequently, the model was further validated using the dataset reported by Yang et al. (Fig. 5b and Table S5) [10] and by Li et al. [11]. At a mixing frequency of 1/d, a substrate concentration of 60 g COD/L, and

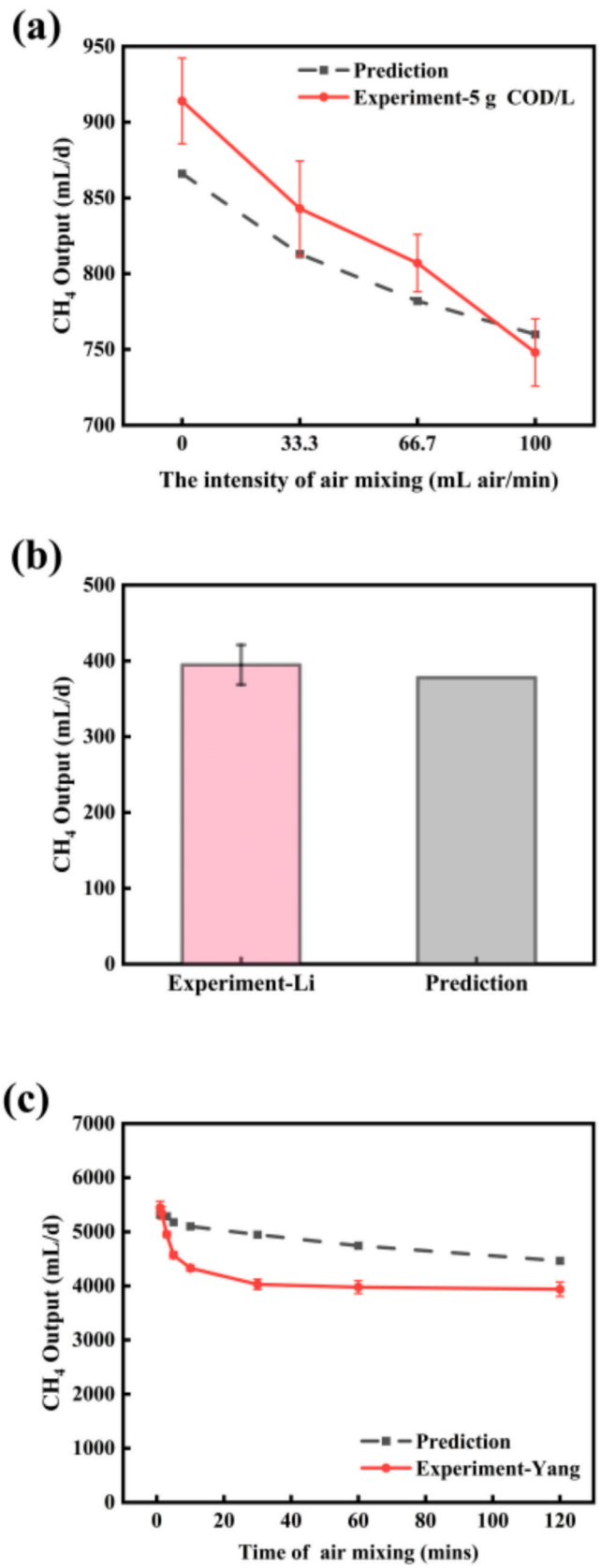


Fig. 5. Comparison between predicted and experimental methane outputs for three independent datasets: (a) an independent 5 g COD/L dataset under different air-mixing intensities, (b) Yang et al. [10] and (c) Li et al. [11].

an intensity of 66.7 mL/(L·min), the methane output decreased with increasing mixing time (from 1 to 120 min). When the mixing time was 1 min, the predicted methane output (5446 mL) was generally consistent with the measured output (5447 mL), with an error of less than 1%. At mixing times of 2–3 min, the predicted output was higher than the measured values by 1.63–8.61%. The error between the predicted and measured values increased further (approximately 14.7%) when air mixing times was longer than 3 min. Yang et al. [10] also reported that mixing times less than 3 min favoured higher methane outputs than no–air mixing did. The increase in mixing time resulted in a rapid decrease in methane output. Li et al. [11] conducted dry anaerobic digestion using cattle manure as substrate and reported that air mixing with twice per day for 1 min each time effectively enhanced methane production. Under these conditions, the methane output reached approximately 394.83 mL/d. Predicting using the model of methane output, a methane output of 378 mL/d for the same conditions can be obtained with a relative error of 4.26% between the predicted and measured values. Overall, the model generated good prediction results, with error values of less than 15% at the air mixing times used in actual anaerobic digestion plants.

Furthermore, Eq. (10) was used to generate Fig. 6, which reflects the effects of mixing time and frequency on methane output at 3 mixing intensities and 3 substrate concentrations. Under an organic substrate concentration of 5 g COD/L and a mixing time of 1 min, when the air mixing intensity was 33 mL air/min, air mixing with a frequency of 4 times/day resulted in a maximum methane output of 984 mL, which was 13.6% higher than that of the control (no air mixing, 866 mL/d). Under the same conditions, when the air mixing intensity was increased to 66.7 and 100 mL/min, the highest methane output reached 1016 and 1022 mL/d (a frequency of 3 times/d), which represented increases of 17.4 and 18.0%, respectively, compared with that of the control. Under the substrate concentration of 5 g COD/L, a mixing intensity of 100 mL air/min, and a mixing time of 1 min, methane output at a frequency of 4 times/d resulted in the maximum value (1022 mL). Under the same substrate concentration and air mixing intensity, the methane output decreased with increasing mixing time. As the mixing frequency increased, the methane output first increased and then decreased.

Under an air mixing intensity of 66.7 mL air/L and an organic substrate concentration of 2 g COD/L, increase of the air mixing time and frequency both resulted in a decrease in methane output. When the organic substrate concentration was increased to 6 and 10 g COD/L, an increase in the mixing frequency increased the methane output. At a organic substrate concentration of 6 g COD/L, mixing for 1 min and 4 times/d resulted in a maximum methane output of 1149 mL/d. When the organic substrate concentration was 10 g COD/L, the methane output reached a maximum of 1785 mL/d at mixing time of 1 min and a frequency 4 times/d. The model further indicates that increasing the organic substrate concentration can offset the negative effects result from increase of mixing frequency, ultimately leading to an effective enhancement in methane output.

Prolonged air mixing time caused the ORP variation and even the irreversible damage to the methanogens, which were the major reasons for the decrease in output [10]. However, increasing the substrate concentration improved the stability of the system significantly. Increasing mixing frequency consistently improves mass transfer efficiency. However, the resulting methane output still depends on substrate concentration and mixing time.

3.5. Implication

Although air mixing can significantly enhance methane output, excessive mixing time or overly frequent mixing may negatively affect anaerobic digestion performance. Therefore, identifying appropriate air-mixing parameters is essential for improving process efficiency in full-scale anaerobic digestion systems. However, experimentally determining the optimal combination of organic substrate concentration,

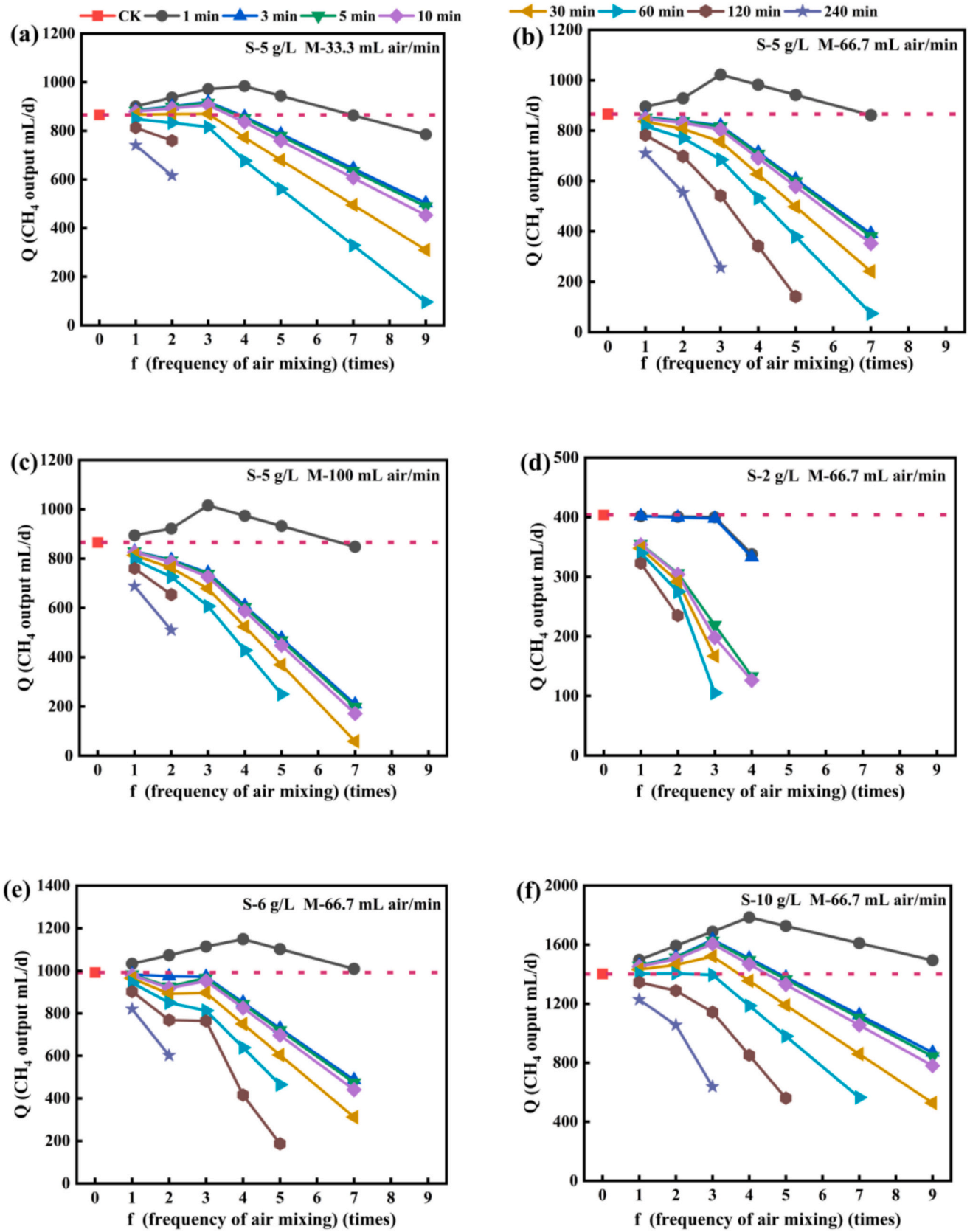


Fig. 6. Effects of mixing time and mixing frequency on methane output under air-mixing conditions. Predictions were performed at an organic substrate concentration of 5 g COD/L with air-mixing intensities of 33.3 mL air/min (a), 66.7 mL air/min (b), and 100 mL air/min (c). Under air-mixing intensity of 66.7 mL air/min, predictions were conducted at organic substrate concentrations of 2 g COD/L (d), 6 g COD/L (e), and 10 g COD/L (f).

mixing intensity, time, and frequency would require an impractically large number of experiments. The kinetic models developed in this study enable prediction of methane output under different organic substrate concentrations and air mixing parameters, thereby allowing direct optimization of mixing intensity, time, and frequency for full-scale engineering applications. The methane output model and the optimized mixing parameters obtained in this study can also be scaled up to full-scale biogas plants. For example, in a biogas plant using the swine wastewater with an organic substrate concentration of 6 g COD/L as feedstock, the optimized mixing parameters can be obtained as follows: air mixing intensity of 0.0166 $m^3/(m^3 \cdot min)$, mixing time of 1 min, and mixing frequency of 4 times per day. Under this condition, the volumetric methane production rate in the digester with air mixing can reach 0.287 $m^3/(m^3 \cdot d)$ as agaist 0.248 $m^3/(m^3 \cdot d)$ in in the digester with no mixing. For the digester with air mixing, each cubic metre of reactor volume thus generated an additional 0.035 m^3 of methane per day. Based on a methane calorific value of 35.9 MJ/m^3, this corresponds to an energy gain of approximately 1.25 $MJ/(m^3 \cdot d)$. For such air mixing intensity, the specific power demand of 26.7 W/m^3 (comparable to conventional mixing practice [44]) multiply by a total daily mixing time of 4 min, resulting in the electrical energy input of about 6.40 $kJ/(m^3 \cdot d)$. The energy output still exceeds the energy input with a output-input ratio of roughly 195, indicating that the energy consumption of air mixing is negligible compared with the additional methane energy recovered. These facts indicate that air mixing has a remarkable energetic advantage.

Overall, the development of these kinetic models provides a quantitative basis for refining and optimizing air mixing–anaerobic digestion technologies, supports the design and operation of full scale biogas plants, and offers new insights into methane emission behavior under micro oxic/anaerobic alternating environment.

4. Conclusions

This study developed kinetics models quantitatively describing ORP variations and methane production, which revealed that high organic substrate concentrations fundamentally enhance the resistance of air mixing–anaerobic digestion systems to oxygen stress. Elevated substrate availability accelerates oxygen consumption and promotes rapid restoration of anaerobic environment, thereby shortening oxygen inhibited periods and stabilizing methanogenic activity after air-mixing. The kinetic models developed demonstrate that the decrease in ORP showed a positive correlation with both the organic substrate concentration and the maximum ORP reached at the end of air mixing. The maximum ORP increase and decrease rates reached 66.0 and 3.03 mV/min, respectively. Methane production followed abnormal, promotion, and normal phases, with a maximum promotion phase rate of 358 mL/h under sufficient substrate availability. The methane output model predicted daily methane production with errors generally below 15%.

Compared with complex models such as ADM1, the proposed ORP based kinetic approach provides a simpler and more practical model for analyzing oxygen stress and optimizing air-mixing–anaerobic digestion systems. The results provide direct guidance for biogas plant design and operation applying the process of air mixing–anaerobic digestion. Given the substrate concentration, the mixing parameters can be determined according to the developed models. This work demonstrates that limited oxygen exposure can be quantitatively managed rather than avoided, offering new insights for the design and control of anaerobic digestion systems.

CRediT authorship contribution statement

Kailun Song: Funding acquisition, Formal analysis, Data curation, Conceptualization. **Huan Chen:** Software, Resources, Methodology. **Yue Zhao:** Investigation, Funding acquisition, Formal analysis. **Li Chen:** Methodology, Investigation, Formal analysis. **Rong You:** Investigation, Formal analysis, Data curation. **Yilin Wang:** Writing – original draft, Visualization, Validation. **Hongnan Yang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision. **Wenguo Wang:** Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation. **Liangwei Deng:** Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2026.172771.

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